

The elemental composition of a sample of a manganese-brass alloy is shown in the table below.

Alloy component	Mass percent
Cu	88.0%
Zn	6.0%
Mn	4.0%
Ni	2.0%

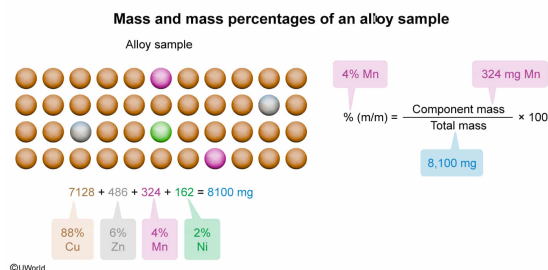
If the sample of the alloy contains 324 mg of Mn, what is the total mass of the sample?

- ☐ A. 810 mg [4%]
☐ B. 7,128 mg [8%]
☐ C. 7,776 mg [13%]
☒ D. 8,100 mg [74%]

Correct

74% answered correctly

Explanation:



Percentages can be used to express the composition of a pure substance or a mixture; however, a variety of different ratios can be chosen as the basis for calculating the percentages. When a mass ratio is used to calculate the percentage of a component, the result is called a **mass percent (% m/m)**, which is defined as the **component mass divided by the total mass** of the substance **expressed as a percentage**:

$$\% (m/m) = \frac{\text{Component mass}}{\text{Total mass}} \times 100$$

In an alloy (ie, a mixture of metals melted together), the **total mass of the alloy** is the **combined mass** of the component metals. Substituting the given mass of Mn and its mass percent from the table into the equation gives:

$$4.0 \% = \frac{324 \text{ mg}}{\text{Total mass}} \times 100$$

Solving for the total mass gives:

$$\frac{4.0 \%}{100} = \frac{324 \text{ mg}}{\text{Total mass}}$$

$$(0.040)(\text{Total mass}) = 324 \text{ mg}$$

$$\text{Total mass} = \frac{324 \text{ mg}}{0.040} = 8,100 \text{ mg}$$

(Choice A) A mass of 810 mg is obtained by incorrectly converting 4.0% to its mass ratio (ie, $4.0/100 \neq 0.40$) in the first step of the calculation.

(Choice B) A mass of 7,128 mg is the mass of Cu in the sample rather than the total mass of the sample (ie, the mass of all metal components).

(Choice C) A mass of 7,776 mg is the combined mass of all metals except Mn in the sample (ie, $8,100 \text{ mg} - 324 \text{ mg} = 7,776 \text{ mg}$). The total sample mass of the sample is the combined mass of *all* the composite metals.

Educational objective:

The mass percent (% m/m) of a component in a sample of a substance or mixture is defined as the component mass divided by the total mass of the sample expressed as a percentage. The total mass of a sample equals the combined mass of each component.

Time spent:0 sec

Subject:
General Chemistry

QID:403954

System:
Atomic Theory and Chemical Composition